# Search provider comparison

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For Priya and the Hal team

## Summary

Exa FT addresses the original evaluation's main weaknesses through explicit location context, mode-specific search and smaller, relevant excerpts. The recorded results are encouraging: **4.53/5** for retrieved evidence, **34 of 40** scores at 4 or 5, and an estimated **$332k monthly cost**.

This supports moving Exa into production-LLM validation for Plan. It does not yet prove that final Hal answers outperform Vendor B: the two score sets use different evaluation methods.

## Why we ran this

Hal already uses Vendor B. Before renewing for another year, Priya asked Wei to compare alternatives using Hal traffic. This report extends Wei's original comparison with the Exa FT work completed on 24 October, with Plan readiness for 1 December as the decision context.

## What we tested

| Provider | Quick questions | Plan | Run date |
|---|---|---|---|
| Vendor B | Search; country set; 5 snippets | Research API | 1 Oct |
| Vendor C | Search; country set; 6 results | Answer API | 8 Oct |
| Exa original | Default search; full text; 10 results | Default search; full text; Hal writes answer; 10 results | 23 Oct |
| Exa FT | Lite; location + highlights; news category where applicable; Chat: 5 results, Voice: 4 results | Deep; objective + highlights; 10 results | 24 Oct |

40 questions: 28 quick (Chat + Voice) and 12 Plan; country and city supplied in the queries tab. FT means API configuration changes, not model-weight fine-tuning.

## Read the numbers correctly

**Quality:** baseline scores assess Hal answers; FT scores assess retrieved evidence. **Timing:** baseline is client-timed search-call latency; FT is Exa server searchTime. **Configuration:** Exa FT shows the agreed setup; saved-run settings differ (page 6). These distinctions apply to every comparison below.

# Recorded scores, speed and context

| Metric | Vendor B | Vendor C | Exa original | Exa FT |
|---|---|---|---|---|
| Average score (all 40) | 3.75 | 3.33 | 2.95 | **4.53** |
| Quick score (28) | 3.75 | 3.39 | 3.00 | **4.82** |
| Plan score (12) | 3.75 | 3.17 | 2.83 | **3.83** |
| Scores of 4 or 5 | 72.5% | 32.5% | 40.0% | **85.0%** |
| Scores of 1 or 2 | 3 | 3 | 20 | **2** |
| Quick search p50 | 651 ms | 687 ms | 1267 ms | **355 ms** |
| Quick search p95 | 1084 ms | 959 ms | 2431 ms | **392 ms** |
| Voice search p50 | 683 ms | 661 ms | 1337 ms | **345 ms** |
| Plan search p50 | 37.8 s | 6.1 s | **1.9 s** | 12.1 s |
| Raw tokens / quick search | 2,355 | **1,921** | 22,263 | 3,039 |
| Cost / quick search | $0.006 | $0.005 | $0.007 | **$0.004** |
| Cost / Plan request | $0.050 | $0.020 | **$0.007** | $0.012 |

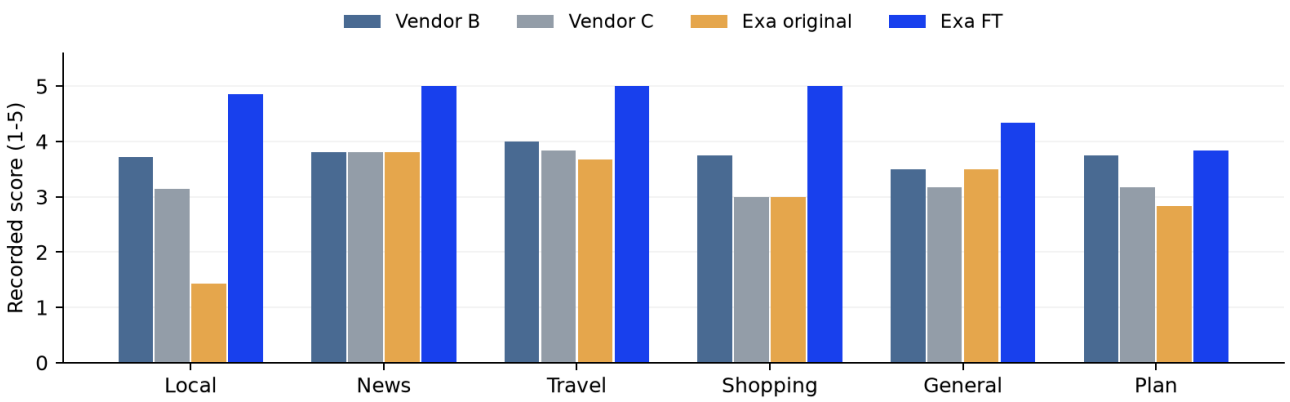
**Bold = best recorded value in each row, including ties.** Higher scores are better; lower latency, token counts and costs are better. Methods differ as noted on page 1.
p50 is the median; p95 is the inclusive percentile across questions, not a repeated-run latency distribution. One recorded result per question; no cold-cache guarantee. Rounded values may differ slightly from the original PDF.

## What changed

**Location:** nearby queries now include city/country text and supplied or resolved coordinates. Explicit destinations take priority over device location.
**Mode:** Lite serves quick turns; Deep handles multi-part Plan requests with an objective.
**Context:** highlights replace full-page text; the retained excerpt budget stays at 6,000 estimated tokens.
**Category:** news intent maps to `category: "news"`; unrelated intents are left unfiltered.

The original Exa quick-search payload averaged 22,263 raw tokens. FT averages 3,039, an 86% reduction. That is a payload improvement; it is not the same as an 86% reduction in model input, because the original input was already capped.

# Where the gains appear




| Intent | Questions | Vendor B | Vendor C | Exa original | Exa FT |
|---|---|---|---|---|---|
| Local | 7 | 3.71 | 3.14 | 1.43 | **4.86** |
| News | 5 | 3.80 | 3.80 | 3.80 | **5.00** |
| Travel | 6 | 4.00 | 3.83 | 3.67 | **5.00** |
| Shopping | 4 | 3.75 | 3.00 | 3.00 | **5.00** |
| General | 6 | 3.50 | 3.17 | 3.50 | **4.33** |
| Plan | 12 | 3.75 | 3.17 | 2.83 | **3.83** |

## Examples from the saved judge notes

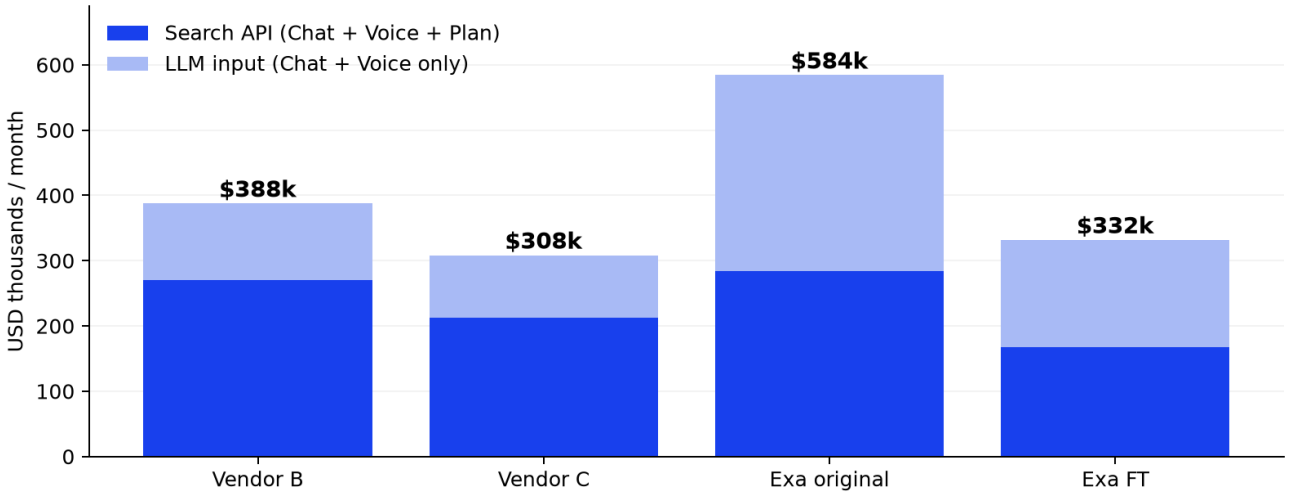
| Query | Original Exa → FT | Recorded finding |
|---|---|---|
| q03 · Bangsar brunch | 1 → 5 | Wrong-city results replaced with Bangsar cafes and local details. |
| q26 · Passport renewal | 1 → 5 | US guidance replaced with Singapore ICA renewal evidence. |
| q32 · Japan tourist visa | 2 → 5 | FT evidence covers JVAC, processing and a document checklist. |
| q30 · HDB grants | 2 → 3 | Loan terms improve; income-specific EHG evidence still missing. |
| q31 · KL to Orchard | 2 → 2 | Missing bus price/time prevents a complete three-way comparison. |

## Remaining gaps

Plan still has five scores below 4. q28 asks to set a reminder, which needs an application action rather than web search. Expected answers were written on 18 September; time-sensitive disagreements require review before comparing judges. The notes above are reported judgments, not independently verified current advice.

# Monthly cost at Hal's volume

Search API costs cover Chat, Voice and Plan. LLM input costs cover retrieved context for Chat and Voice only; Plan model input is excluded.




| Monthly cost (USD) | Vendor B | Vendor C | Exa original | Exa FT |
|---|---|---|---|---|
| Search API (Chat + Voice + Plan) | $270,000 | $212,000 | $284,200 | **$167,200** |
| LLM input (Chat + Voice only) | $117,611 | **$96,360** | $299,956 | $164,645 |
| Total | $387,611 | **$308,360** | $584,156 | $331,845 |

## Exa FT: about $56k/month lower than Vendor B

That is **14.4% lower modeled cost** at the stated volume. Search spend falls substantially, while FT's quick-answer context still costs more than Vendor B's model input. Vendor C remains the lowest-cost option in this model.

Calculation: 30M × mean Chat cost + 10M × mean Voice cost + 600k × mean Plan cost. Model input: (30M × mean Chat tokens used + 10M × mean Voice tokens used) × $1.25 / 1M.

Uses recorded per-call prices and saved tokens, not a new price quote. Excludes Plan LLM input, all model output, voice generation, geocoding, retries and enterprise discounts. The current 5 / 4 / 10 setup needs a fresh run before projecting its exact cost.

# Chat and Voice requests

JavaScript / Exa SDK examples. Supply query and verified user location at runtime; keep local city/country text in the query. Omit category unless it maps to a supported result category.

```
import Exa from "exa-js";
const exa = new Exa(process.env.EXA_API_KEY);

// Resolve location before search; do not invent coordinates.
const userLocation = { country, latitude, longitude };
const localQuery = `${query}\nUser location: ${city}, ${countryName}.
Respect any destination explicitly named in the request.`;
const filter = intent === "news" ? { category: "news" } : {};
```

## Chat · Lite · 5 results

```
const chatResults = await exa.search(localQuery, {
  type: "lite",
  numResults: 5,
  userLocation,
  ...filter,
  contents: { highlights: true }
});
```

## Voice · Lite · 4 results

```
const voiceResults = await exa.search(localQuery, {
  type: "lite",
  numResults: 4,
  userLocation,
  ...filter,
  contents: { highlights: true }
});
```

The result count changes by mode; highlights stays true. Search latency is measured around the Exa call independently of speech generation. The voice provider does not need to finish before search metrics can be recorded.

# Plan request

## Plan · Deep · 10 results

```
const planResults = await exa.search(localQuery, {
  type: "deep",
  numResults: 10,
  userLocation,
  ...filter,
  objective, // User constraints and required evidence
  contents: { highlights: true }
});
```

## Sources and reproducibility

Original baseline: *Search provider comparison - Oct 2026.pdf*, Wei Lim, 23 Oct 2026. Live sheet: results_exa_fine_tune, queries and four results tabs. Current setup: hal/lib/search-profiles.mjs and hal/lib/context-budget.mjs. Samples: outputs/highlights-three-modes/ and outputs/highlights-plan-q32/. API detail: Exa highlights documentation.

Saved-run provenance: the sheet notes earlier Chat searches used 6 results, despite later setup labels showing 5. Saved Plan rows show dynamic highlights and 8 results; Voice uses 4. Some Plan rows omit coordinates. The 40-row results must not be presented as a fresh all-true, 5 / 4 / 10 benchmark.

# Evaluation script

Saved search results → Gemini → score and judge note

## 1. Open the files

**hal/scripts/judge.mjs**
Loads saved results, calls Gemini and saves judgments. Completed records are skipped when you resume the same run.

**hal/scripts/evaluation/judge-core.mjs**
Defines the scoring rubric, prepares the evidence and validates the returned score.

**/path/to/search-results.json**
Supply your own exported search results; replace this generic path with your file. Raw evaluation results are not included. The script does not rerun Exa.

## 2. Run a small check

Set `GEMINI_API_KEY` and `GEMINI_JUDGE_MODEL` in `hal/.env.local`. From the project folder, run:

```
cd hal
npm run judge -- \
  --input /path/to/search-results.json \
  --output .data/evaluation/retrieval.json \
  --dry-run
```

**Dry run:** validates inputs without calling Gemini.
**Run judging:** remove `--dry-run`.
**Selected queries:** optionally add `--ids q01,q02,q03` using IDs from your input.
**Resume:** repeat the command after an interruption.

## 3. Read the result

Open `hal/.data/evaluation/retrieval.json`. Review `score` (1-5) and `judge_note` alongside the evidence. Higher scores indicate better coverage of the expected criteria.

**What this measures:** retrieved evidence, not necessarily the final Hal answer. To judge final answers, supply records containing the actual answers and use `--target answer`. Automated judgments still need review; they do not establish production performance or citation accuracy on their own.